Supplemental Material for "Improved first-principles equation-of-state table of deuterium for high-energy-density science applications"

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Here, we present details about calculations at $\rho = 0.001$ g/cm^3 which demonstrate the accuracy of OFMD in the low-density, high-temperature regime in Sec. 1. Also, we compare the iFPEOS interpolation region with the REOS.3 table, which is designed to target this low-energy-density regime. The full iFPEOS table on the density-temperature points as shown in Fig. 2 in the manuscript is provided in Sec. 2.

1 Calculations at low densities.

For densities below $\rho = 0.1$ g/cm^3 we perform low-temperature, Kohn-Sham molecular dynamics (KSMD) calculations for the lowest-density isochore $\rho = 0.001$ g/cm^3 only, due to the computationally expensive nature of KSMD calculations in this regime. For density points $0.001 < \rho < 0.01$ g/cm^3, $T < 125000$ K we interpolate based on: (1) our KSMD/OFMD results for $\rho = 0.001, 0.1, 0.1996$ and 0.3065 g/cm^3 isochores from $T = 800$ K to $T = 500$ kK and (2) the OFMD results for the $T = 181\ 825$, 250 000, 400 000, 500 000 K isotherms from $\rho = 0.001$ g/cm^3 to $\rho = 0.3065$ g/cm^3. Particularly challenging are the calculations for the lowest-density isochore $\rho = 0.001$ g/cm^3 due to the large box size and the rapidly growing number of thermally occupied bands. The highest temperature at which KSMD calculations were possible is $T < 125000$ K. At this temperature, OFMD calculations were performed and results were compared to those from KSMD calculations. We find that OFMD overestimates the pressure by 1.7%. Here we note that we perform OFMD calculations with LKTFγTF, $\gamma = 0.233$, as consistent with our analytical fit for the tunable γ parameter (See Fig. 3 in main text), however at these conditions we see a weak dependence on the γ parameter. At lower temperatures the disagreement between OFMD and KSMD grows (See Figs. S1 and S3 below), which is expected as OFMD becomes less accurate. This convergence between OFMD and KSMD at high temperatures serves as rationale for using OFMD for $T \geq 181\ 825$ K where we expect accuracy of OFMD calculations to be within $\sim 1\%$.

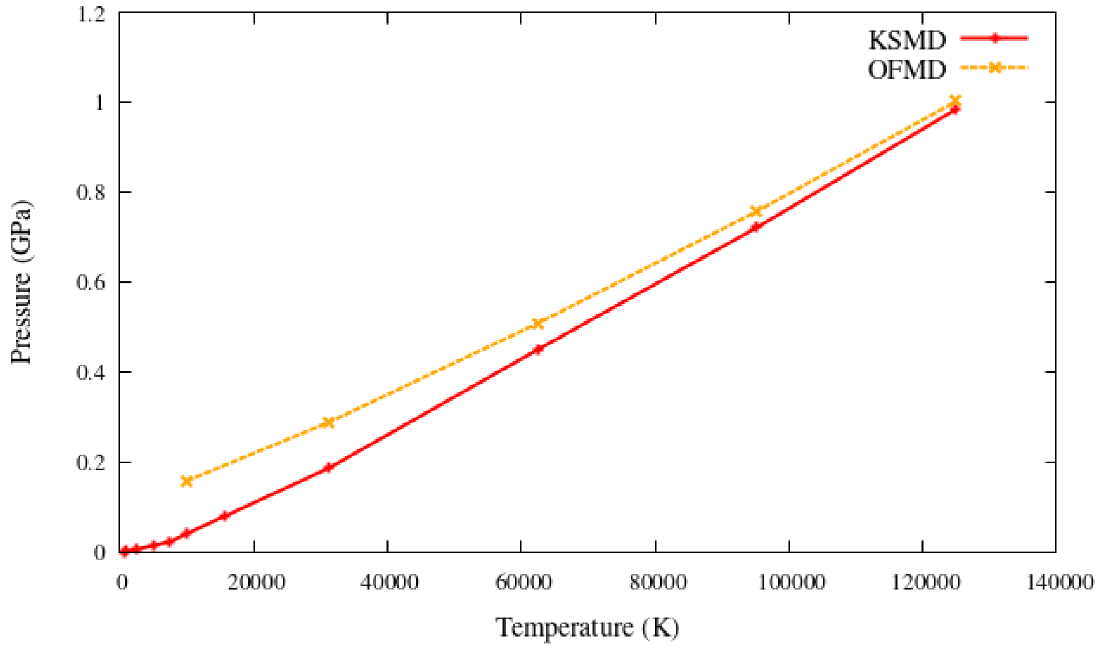

Figure S1: OFMD (green) and KSMD (purple) results for pressure along the $\rho = 0.001$ g/cm^3 isochore for temperatures above 15 625 K. At lower temperatures OFMD is even less accurate as it does not capture the D-D bond. At the next T-point along this isochore, $T = 250$ 000 K, it is expected that results for pressure obtained with the two methods will agree within $\sim 1\%$.

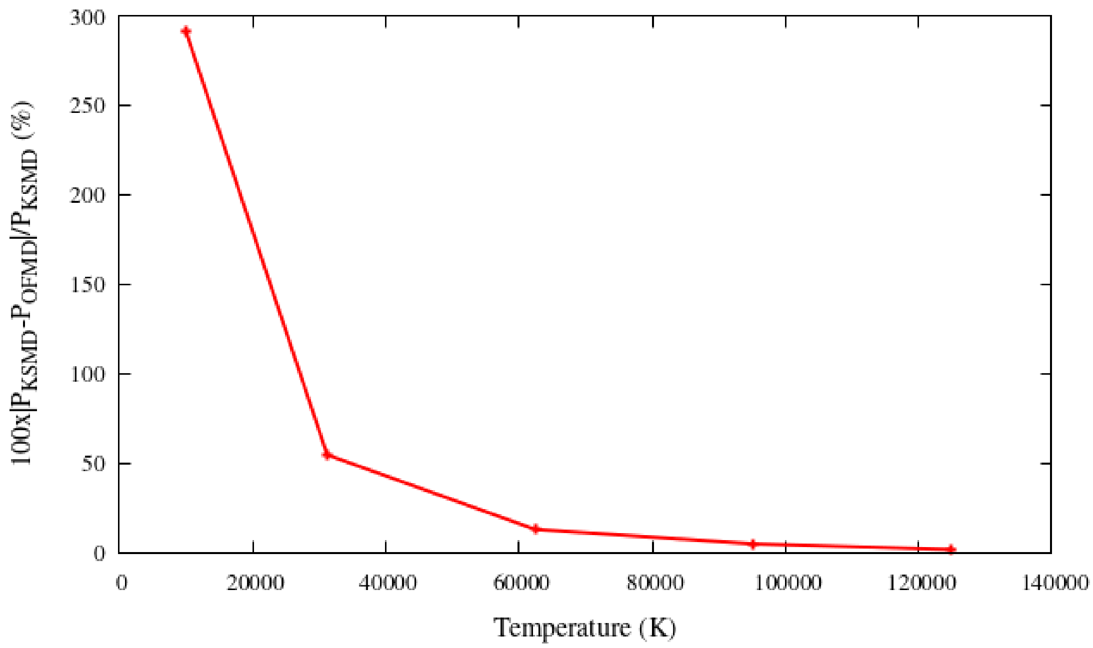

Figure S2: Relative difference in pressures between OFMD and KSMD along the $\rho = 0.001$ g/cm^3 isochore. At $T = 125000$ K (the highest temperature at which KSMD was done) the difference is only 1.7%.

For the regime $0.002 \leq \rho \leq 0.084$ g/cm^3, 800 K$\leq T \leq 182$ kK we use spline interpolation using our results for the $\rho = 0.001, 0.1, 0.3$ and 0.38 g/cm^3 isochores from $T = 800$ K to $T = 500$ kK, and the $T = 182, 250, 400, 500$ kK isotherms from $\rho = 0.001$ g/cm^3 to $\rho = 0.3$ g/cm^3 (orange circles in lower left quadrant of Fig. 2 in the main text). Below is a comparison between of the iFPEOS low-density region with that from the well-established REOS.3 table [13].

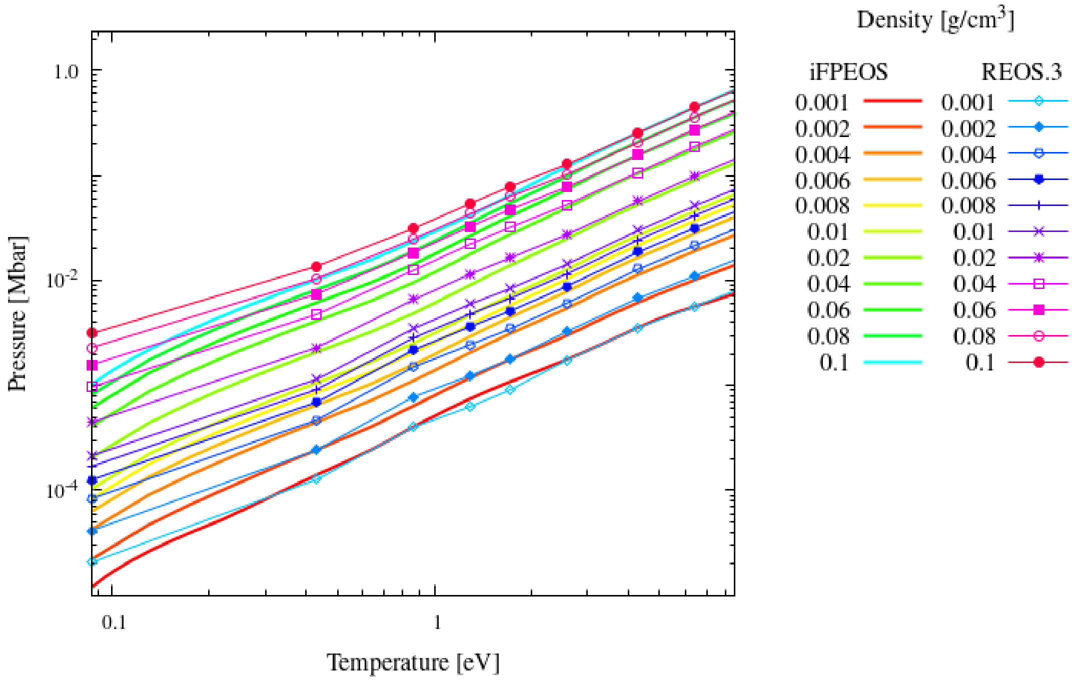

Figure S3: Pressure as a function of temperature for the lowest-density isochores according to iFPEOS (lines) and REOS.3 [13] (lines with points). The $\rho = 0.1$ g/cm^3 and The $\rho = 0.001$ g/cm^3 are results from direct KSMD calculations. The rest of the iFPEOS isochores shown here have been obtained through spline interpolation [78]. Note that these particular isochores have been separately obtained in order to make a direct comparison with REOS.3 and do not actually appear in the table provided below.

2 iFPEOS table

Table S1: Temperature, pressure and internal energy for each density point. The uncertainty reported is the standard deviation during the MD run. Standard deviation of 0.0 corresponds to results obtained by interpolation.

$\rho = 0.001000$ g/cm^3		
Temperature (K)	Pressure (Mbar)	Internal Energy (eV/atom)
800	0.000006 ± 0.000022	-15.680 ± 0.003
900	0.000009 ± 0.000000	-15.667 ± 0.000
1000	0.000012 ± 0.000028	-15.654 ± 0.005
1100	0.000015 ± 0.000000	-15.641 ± 0.000
1200	0.000018 ± 0.000000	-15.628 ± 0.000
1300	0.000021 ± 0.000000	-15.614 ± 0.000
1400	0.000023 ± 0.000000	-15.601 ± 0.000
1500	0.000026 ± 0.000000	-15.588 ± 0.000
1600	0.000029 ± 0.000000	-15.575 ± 0.000
1700	0.000031 ± 0.000000	-15.561 ± 0.000
1800	0.000034 ± 0.000000	-15.548 ± 0.000
1900	0.000036 ± 0.000000	-15.535 ± 0.000
2000	0.000039 ± 0.000000	-15.522 ± 0.000
2100	0.000041 ± 0.000000	-15.508 ± 0.000
2200	0.000044 ± 0.000000	-15.495 ± 0.000
2300	0.000047 ± 0.000000	-15.482 ± 0.000
2400	0.000049 ± 0.000000	-15.468 ± 0.000
2500	0.000052 ± 0.000061	-15.455 ± 0.010
5000	0.000141 ± 0.000039	-15.138 ± 0.007
7500	0.000249 ± 0.000241	-13.235 ± 0.008
10000	0.000403 ± 0.000044	-12.005 ± 0.008
15625	0.000790 ± 0.000000	-8.847 ± 0.000
31250	0.001863 ± 0.000471	-1.384 ± 0.153
62500	0.004701 ± 0.000835	12.951 ± 0.096
95250	0.007159 ± 0.000033	20.845 ± 0.128
125000	0.009842 ± 0.000019	28.883 ± 0.068
181825	0.014808 ± 0.000042	45.698 ± 0.260
250000	0.020460 ± 0.000036	63.484 ± 0.227
400000	0.032876 ± 0.000045	102.468 ± 0.279
500000	0.041137 ± 0.000048	128.354 ± 0.300
1000000	0.082444 ± 0.000046	257.777 ± 0.287
2000000	0.165027 ± 0.000059	516.425 ± 0.369
4000000	0.330152 ± 0.000048	1033.473 ± 0.299
8000000	0.660400 ± 0.000050	2067.545 ± 0.313
16000000	1.320904 ± 0.000055	4135.746 ± 0.345
32000000	2.641888 ± 0.000043	8271.995 ± 0.270
64000000	5.283894 ± 0.000053	16544.729 ± 0.334
128000000	10.567885 ± 0.000055	33090.062 ± 0.343
256000000	21.135864 ± 0.000048	66180.641 ± 0.302